

Arpan Pathak

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SUMMARY

Senior Software Engineer with **8 years** designing high-scale distributed systems, real-time ML platforms, and cloud infrastructure at **Amazon, Microsoft, and Oracle**. Deep in **Rust, C++, Java, Kotlin, Go, Python**: low-latency services, lock-free concurrency, event-driven pipelines (Kafka/Spark), and production ML inference (XGBoost, PyTorch, BERT) at global scale.

EXPERIENCE

Software Developer II Microsoft — Redmond, WA Jun 2025 – Present

- **C++ → Rust Migration**: Rebuilt Microsoft Information Protection from **legacy C++/C#** in **Rust** — compile-time memory safety, eliminating use-after-free, buffer-overflow, and null-safety bugs; **~40% lower P99 latency, 2x throughput, 3x lower memory** per request, 6-figure annual savings.
- **eBPF/Cilium Networking**: Architected zero-trust network policies for sovereign cloud (data residency, SecNumCloud); built **Kubernetes operators** (Kubebuilder) reconciling policy state across AKS clusters in France and Germany.
- **AI Policy Engine**: Auto-generates Cilium policies and scans misconfigurations — manual review cut from **4 hours** to near-real-time.

Senior Software Engineer Oracle Cloud Infrastructure — Seattle, WA Oct 2024 – Mar 2025

- **Database Internals**: Optimized **lock-free data structures** in a proprietary database engine — **+45,000 ops/s** throughput in high-concurrency read-write paths.
- **Terraform SDK**: Built the Terraform provider for Oracle Autonomous Database, replacing legacy control-plane tooling — **\$1.2M/month** projected savings.

Software Development Engineer II Amazon — Seattle, WA & Hyderabad, India Mar 2021 – Oct 2024

- **Real-Time ML Ranking**: **XGBoost + deep-learning ensemble** ranking ads/deals across FreeVee, Twitch, Prime Video, and Amazon Retail — **100ms inference** on millions of events, **5% CTR lift, \$5.4M/month revenue growth**.
- **BERT NLP Service**: **5,000 QPS** at peak; cut support-ticket triage SLA from **7 days** to **2 hours**.
- **NoSQL SDK (two-phase commit)**: Uniqueness-constraint indexing SDK for a multi-tenant store — **serializable isolation** at **3K writes/s**.
- **Data Platform**: **1-petabyte** warehouse/lake with **Spark (AWS Glue)** pipelines for financial reporting and BI.

SDE Razorpay — Go UPI gateway (1M+ hourly txns, 99.99% uptime) & **Mindfire** — Java/Spring full-stack, AR search 2018 – 2021

SELECTED PROJECTS

Driving-CivicSense — On-Device Vision (Rust) | github.com/arpanpathak/driving-civicsense-vision-model

- **YOLOv8/v11** on low-power ARM boards at **<50ms** on-device inference; author of **Seeing Machines** & **CUDA Kernels** books.

SKILLS

Languages: Rust, C++, C, Go, Java, Kotlin, Python, TypeScript

Systems & Performance: Lock-free concurrency, memory safety, eBPF, performance engineering, Linux

Data, Cloud & ML: Kafka, Spark, DynamoDB, PostgreSQL, Redshift, AWS, Azure, GCP — PyTorch, XGBoost, BERT, ONNX, edge inference

EDUCATION

B.Tech, Computer Science & Engineering — RCC Institute of Information Technology, Kolkata | 2018